Chapter 2 solutions

CH02-E1

Subtract twice the first equation from the second.

$$(2x - y) - 2(x + 3y) = 0 - 14,$$

Thus $-7y = -14$, so $y = 2$. Substitution gives $x + 6 = 7$, hence $x = 1$.

In the row picture, the lines intersect at $(1, 2)$.

In the column picture,

$$1\begin{bmatrix}1\\2\end{bmatrix} + 2\begin{bmatrix}3\\-1\end{bmatrix} = \begin{bmatrix}7\\0\end{bmatrix}.$$

The intersection coordinates are also the column coefficients.

CH02-E2

The first pivot is $A_{11} = 1$. Use multiplier $3/1 = 3$ to clear the first entry of row 2.

$$\begin{bmatrix}1 & 2 & 1\\0 & 2 & 1\\0 & 4 & 3\end{bmatrix}.$$

The second pivot is 2. Use multiplier $4/2 = 2$ to clear the second entry of row 3.

$$\begin{bmatrix}1 & 2 & 1\\0 & 2 & 1\\0 & 0 & 1\end{bmatrix}.$$

The third pivot is 1. Three nonzero pivots give rank 3, so A is invertible.

CH02-E3

The shapes are 2×3 for A and 3×2 for B. The inner dimensions match, so

$$AB \in \mathbb{R}^{2\times 2}.$$

Compute

$$AB = \begin{bmatrix}1\cdot 2 + 2\cdot 1 + 0\cdot 0 & 1\cdot 1 + 2\cdot 0 + 0\cdot(-1)\\0\cdot 2 + 1\cdot 1 + 3\cdot 0 & 0\cdot 1 + 1\cdot 0 + 3\cdot(-1)\end{bmatrix} = \begin{bmatrix}4 & 1\\1 & -3\end{bmatrix}.$$

The reverse product also exists, with shape 3×3.

$$BA = \begin{bmatrix}2 & 5 & 3\\1 & 2 & 0\\0 & -1 & -3\end{bmatrix}.$$

The products have different shapes, so they cannot be equal.

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CH02-E4

Begin with

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{array}\right].$$

Subtract row 2 from row 1.

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 1 & 1 & 0 & 1 \end{array}\right].$$

Then subtract the new row 1 from row 2.

$$\left[\begin{array}{cc|cc} 1 & 0 & 1 & -1 \\ 0 & 1 & -1 & 2 \end{array}\right].$$

Therefore

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Verification gives

$$\begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

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CH02-E5

The elimination multiplier is

$$\ell_{21} = \frac{4}{2} = 2.$$

Thus

$$L = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix}.$$

The operation $R_2 \leftarrow R_2 - 2R_1$ gives

$$U = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}.$$

Check

$$\begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix} = A.$$

CH02-E6

The inner product is a 1×1 matrix, identified with a scalar.

$$x^T y = 1 \cdot 3 + 2 \cdot (-1) = 1.$$

The outer product multiplies a 2×1 matrix by a 1×2 matrix, giving shape 2×2.

$$xy^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}.$$

The second column is $-1/3$ times the first. Both are nonzero, and their span is

$$\operatorname{span}\{(3, 6)^T\},$$

which has dimension one. Therefore xy^T has rank one.

CH02-E7

1.

$$G = \begin{bmatrix} \langle p_1, p_1 \rangle & \langle p_1, p_2 \rangle \\ \langle p_2, p_1 \rangle & \langle p_2, p_2 \rangle \end{bmatrix} = \begin{bmatrix} \int_{-1}^{1} 1 \, dt & \int_{-1}^{1} (1+t) \, dt \\ \int_{-1}^{1} (1+t) \, dt & \int_{-1}^{1} (1+t)^2 \, dt \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 8/3 \end{bmatrix}.$$

For nonzero $c = (c_1, c_2)^T$, the Gram form is

$$c^T G c = \int_{-1}^{1} (c_1 + c_2 + c_2 t)^2 \, dt > 0,$$

because a nonzero linear polynomial cannot vanish almost everywhere on the interval. Thus G is positive definite. Also $\det G = 16/3 - 4 = 4/3 > 0$.

2. The right-hand side is

$$b = \begin{bmatrix} \langle p_1, f \rangle \\ \langle p_2, f \rangle \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Gram system：

$$\begin{bmatrix} 2 & 2 \\ 2 & 8/3 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

The first equation gives $c_1 + c_2 = 1$. The second is $2c_1 + (8/3)c_2 = 4$. Substitute $c_1 = 1 - c_2$. Then $2 - 2c_2 + (8/3)c_2 = 4$, so $(2/3)c_2 = 2$, $c_2 = 3$, $c_1 = -2$. Substitution gives $-2 + 3(1 + t) = 1 + 3t$.

3. The three basis functions are pairwise orthogonal, with $\|1\|^2 = 2\pi$, $\|\cos x\|^2 = \|\sin x\|^2 = \pi$. Therefore

$$a_0 = \frac{1}{2\pi} \int_0^{2\pi} (1 + x) \, dx = 1 + \pi,$$

$$a_1 = \frac{1}{\pi} \int_0^{2\pi} (1 + x) \cos x \, dx = 0,$$

$$b_1 = \frac{1}{\pi} \int_0^{2\pi} (1 + x) \sin x \, dx = \frac{-2\pi}{\pi} = -2.$$

Hence $f_1(x) = 1 + \pi - 2 \sin x$.

4. The residual is

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$$e = f - f_1 = x - \pi + 2\sin x.$$

$$\int_0^{2\pi} e(x)\,dx = \left[\frac{x^2}{2} - \pi x - 2\cos x\right]_0^{2\pi} = 2\pi^2 - 2\pi^2 - 2 + 2 = 0,$$

$$\int_0^{2\pi} e(x)\cos x\,dx = 0$$

because both $x\cos x$ and a constant times $\cos x$ integrate to zero over the full period. Also

$$\int_0^{2\pi} e(x)\sin x\,dx = \int_0^{2\pi} (x-\pi)\sin x\,dx + 2\int_0^{2\pi} \sin^2 x\,dx = -2\pi + 2\pi = 0.$$

Thus the residual is orthogonal to the entire finite subspace, precisely the projection normal equations.

CH02-E8

1. The first row of $T_n(a)$ is $(a, 1, 0, \ldots, 0)$. The cofactor multiplying a is D_{n-1}. In the minor associated with the adjacent 1, the next 1 leaves $T_{n-2}(a)$ after the first two rows and columns are removed. Hence

$$D_n = aD_{n-1} - D_{n-2}, \qquad n \geq 2.$$

2.

$$D_1 = a, \qquad D_2 = a^2 - 1,$$
$$D_3 = a(a^2-1) - D_1 = a^3 - 2a = a(a^2-2).$$

Thus $T_3(a)$ is invertible exactly when $a \notin \{-\sqrt{2}, 0, \sqrt{2}\}$.

3. Let $v_j = \sin(j\theta_k)$, with boundary values $v_0 = v_{n+1} = 0$. For $j = 1, \ldots, n$,

$$(T_n(a)v)_j = v_{j-1} + av_j + v_{j+1}.$$

the trigonometric identity $\sin((j-1)\theta) + \sin((j+1)\theta) = 2\cos\theta\sin(j\theta)$ gives

$$(T_n(a)v)_j = (a + 2\cos\theta_k)v_j.$$

The boundary indices also work because $\sin 0 = \sin((n+1)\theta_k) = \sin(k\pi) = 0$.

4. Distinct θ_k give distinct eigenvalues. Symmetry supplies an orthogonal eigenbasis, and the determinant is their product.

$$D_n(a) = \prod_{k=1}^{n} (a + 2\cos(k\pi/(n+1))).$$

For $n = 3$, $\theta = (\pi/4, \pi/2, 3\pi/4)$, with cosines $\sqrt{2}/2, 0, -\sqrt{2}/2$, and eigenvalues $a + \sqrt{2}, a, a - \sqrt{2}$. At $a = 3$,

$$(3 + \sqrt{2})3(3 - \sqrt{2}) = 3(9 - 2) = 21.$$

agreeing with $D_3(3) = 3(9-2) = 21$.

CH02-E9

1. $AB = \begin{bmatrix} 4 & 1 \\ 1 & -3 \end{bmatrix}$ is 2×2, and $BA = \begin{bmatrix} 2 & 5 & 3 \\ 1 & 2 & 0 \\ 0 & -1 & -3 \end{bmatrix}$ is 3×3. Their traces are $4 - 3 = 1$ and $2 + 2 - 3 = 1$, respectively.
2. If A is $m \times n$ and B is $n \times m$, then $\operatorname{tr}(AB) = \sum_{i=1}^{m} \sum_{j=1}^{n} a_{ij} b_{ji} = \sum_{j=1}^{n} \sum_{i=1}^{m} b_{ji} a_{ij} = \operatorname{tr}(BA)$. Only finite sums and scalar products are rearranged.
3. Over the reals, $\operatorname{tr}(AB - BA) = 0$ but $\operatorname{tr}(I_n) = n \neq 0$. In characteristic 2 with $n = 2$, the identity has trace zero, so the contradiction disappears. Indeed $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$ satisfy $AB - BA = I_2$ over that field.

   The products are

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad BA = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \qquad AB - BA = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix},$$

using $-1 = 1$ in this field for the last equality.

CH02-E10

1. The multipliers $\ell_{21} = 2, \ell_{31} = 3$ give

$$\begin{bmatrix} 2 & 1 & 1 \\ 0 & 0 & 1 \\ 0 & 1 & 2 \end{bmatrix}.$$

The second pivot position is zero but the entry below is 1. Swap rows 2 and 3; this is not evidence of singularity.

2. The factors are

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 2 & 0 & 1 \end{bmatrix}, \quad U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

The stored first-column multipliers 2 and 3 must swap because they describe the corresponding original equations. Multiplication gives

$$LU = \begin{bmatrix} 2 & 1 & 1 \\ 6 & 4 & 5 \\ 4 & 2 & 3 \end{bmatrix} = PA.$$

Leaving L_{21}, L_{31} equal to 2 and 3 would reconstruct the wrong equations.

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3. $Pb = (3, 12, 8)^T$. Forward substitution gives $y_1 = 3$, $3y_1 + y_2 = 12 \Rightarrow y_2 = 3$, $2y_1 + y_3 = 8 \Rightarrow y_3 = 2$. Back substitution gives $x_3 = 2$, $x_2 + 2x_3 = 3 \Rightarrow x_2 = -1$, and $2x_1 + x_2 + x_3 = 3 \Rightarrow x_1 = 1$. Check $A(1, -1, 2)^T = (3, 8, 12)^T$.
4. $\det A_1 = 2$, $\det A_2 = 2 \cdot 2 - 4 \cdot 1 = 0$. Since $\det U = 2$ and $\det P = -1$, $\det A = -2 \neq 0$. Invertibility ensures that suitable swaps can find pivots, but does not ensure that every leading principal submatrix is invertible.

References and attribution

Lin Cing, handwritten linear algebra notes. Gilbert Strang, Introduction to Linear Algebra, 5th ed., Chapter 2.

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